JavaScript RegExp Quantifiers

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RegExp Quantifiers

Quantifiers specify how many times a pattern should match:

Quantifier	Description
`+`	Match 1 or more times
`*`	Match 0 or more times
`?`	Match 0 or 1 times
`{n}`	Match exactly n times
`{n,m}`	Match between n and m times
`{n,}`	Match n or more times

RegExp + Quantifier

`+` matches 1 or more occurrences of the preceding character/pattern:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Quantifiers</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp + Quantifier</h4>
<p id="demo"></p>

<script>
const text = "hello world hello!!";
const matches = text.match(/o+/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/o+/g): " + matches + "<br>" +
  "'+' matches one or more 'o' characters";
</script>

</body>
</html>
```

Example 1

Result [View Example]

RegExp * Quantifier

`*` matches 0 or more occurrences of the preceding pattern:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Quantifiers</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp * Quantifier</h4>
<p id="demo"></p>

<script>
const text = "Hello World";
const matches = text.match(/lo*/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/lo*/g): " + matches + "<br>" +
  "'*' matches zero or more 'o' characters after 'l'";
</script>

</body>
</html>
```

Example 2

Result [View Example]

RegExp ? Quantifier

`?` matches 0 or 1 occurrence of the preceding character:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Quantifiers</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp ? Quantifier</h4>
<p id="demo"></p>

<script>
const text = "color colour";
const matches = text.match(/colou?r/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/colou?r/g): " + matches + "<br>" +
  "'?' makes 'u' optional (0 or 1)";
</script>

</body>
</html>
```

Example 3

Result View Example

RegExp {n} Quantifier

`{n}` matches exactly n occurrences:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp Quantifiers</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp {n} Quantifier</h4>
<p id="demo"></p>

<script>
const phone = "555-123-4567";
const matches = phone.match(/\d{3}/g);
const match4 = phone.match(/\d{4}/g);

document.getElementById("demo").innerHTML =
  "Text: " + phone + "<br><br>" +
  "match(/\\d{3}/g): " + matches + " (exactly 3 digits)<br>" +
  "match(/\\d{4}/g): " + match4 + " (exactly 4 digits)";
</script>

</body>
</html>
```

Example 4

Result View Example

Document

Document in project

You can Download PDF file.

Reference

- W3Schools JavaScript RegExp Quantifiers